

An Ultrahigh-Performance Photodetector based on a Perovskite–Transition-Metal-Dichalcogenide Hybrid Structure

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Transition-metal dichalcogenides (TMDs) with a 2D semiconducting structure, such as molybdenum disulfide (MoS_2), molybdenum diselenide (MoSe_2), tungsten disulfide (WS_2), and tungsten diselenide (WSe_2), are considered to be some of the most promising nanomaterials for future electronic and optoelectronic applications^[1–5] due to their exceptional physical,^[6,7] electrical,^[8–10] and optical^[11–13] properties. In particular, because of their extremely high quantum efficiency,^[14] ($\approx 45\%$ for MoS_2), which surpasses that of conventional semiconductors ($\approx 40\%$ for GaAs, $\approx 37\%$ for InP, and $\approx 24\%$ for Si),^[15–17] TMD-based optoelectronic devices have recently shown excellent performance (i.e., photoresponsivity in photodetectors^[11–13] and efficiency in solar cells.^[18,19]). However, despite their superior optical properties, it is physically difficult for TMDs to absorb sufficient light because of their atomically thin thickness,^[20] which thus prevents the successful integration of TMD-based optoelectronic devices. With that consideration, various efforts have been made to improve the characteristics of TMD-based optoelectronics, especially photodetectors, by inserting an absorption layer with a very high absorption rate and quantum efficiency, for example carbon nanotubes,^[21] silicon (Si),^[22] Rhodamine 6G (R6G),^[23] or PbS quantum dots.^[24,25] Jariwala et al.^[21] presented a highly improved photoresponsivity value (0.1 A W^{-1}) in a MoS_2 -based photodetector with carbon-nanotube decoration on

the detection area. Salahuddin and Esmaeili-Rad^[22] also demonstrated a Si/ MoS_2 heterojunction photodetector with a photoresponsivity value of $\approx 210 \text{ mA W}^{-1}$. Yu et al.^[23] applied R6G to an MoS_2 photodetector to improve its absorption property and obtained excellent performance (1.17 A W^{-1} photoresponsivity and 1.5×10^7 Jones detectivity). Kufer et al.^[24] and Konstantatos et al.^[25] used PbS quantum dots for an additional absorption layer on 2D-based photodetectors and respectively reported high photoresponsivity values of 6×10^5 and $1 \times 10^7 \text{ A W}^{-1}$.

Meanwhile, organolead trihalide perovskite materials (MAPbX_3 , where MA is methylammonium CH_3NH_3 and X is a halogen), which are considered next-generation solar-cell materials, were recently applied on photodetectors based on reduced graphene oxide (rGO),^[26] graphene (Gr),^[27] and Gr/ WSe_2 /Gr^[28] and have shown very high photoresponsivity values of 73.9 mA W^{-1} , 180 A W^{-1} and 950 A W^{-1} , respectively. These results were attributed to the long electron–hole diffusion length^[29] and ambipolar charge transport^[30] of the perovskite. However, the high conductivity of the rGO and graphene layers caused very high dark currents and subsequently low detectivity in the photodetectors. Therefore, it is necessary to combine “right” semiconductors (transporting layers), which can suppress a dark current, and an excellent light absorbing layer such as perovskite to achieve excellent photodetector performance. Here, we demonstrate an ultrahigh-performance MoS_2 photodetector with record photoresponsivity and detectivity based on a perovskite/ MoS_2 hybrid structure. Because of the high absorption and long diffusion lengths of charge carriers of the perovskite, many electrons and holes are expected to be generated in the perovskite and then be transported to the MoS_2 . To characterize the perovskite/ MoS_2 hybrid structure, we performed photoluminescence (PL) and rigorous couple-wave analysis (RCWA). We then systematically investigated the perovskite/ MoS_2 photodetector in terms of optoelectronic performance (photoresponsivity, detectivity, and photoswitching) by optical measurements (I_D-V_G and I_D-V_D) under dark and illuminated conditions. Finally, we monitored the stability of the perovskite/ MoS_2 photodetector fabricated on (3-aminopropyl) triethoxysilane (APTES) in air, where we proposed and investigated an encapsulation bilayer of poly(methyl methacrylate) (PMMA) and octadecyltrichlorosilane (OTS) layers to improve stability.

To investigate the characteristics of the perovskite/ MoS_2 hybrid structure, we performed PL and UV–vis analyses on methyl-ammonium lead-halide perovskite/ MoS_2 samples. Figure 1a shows the schematic and energy-band diagram of

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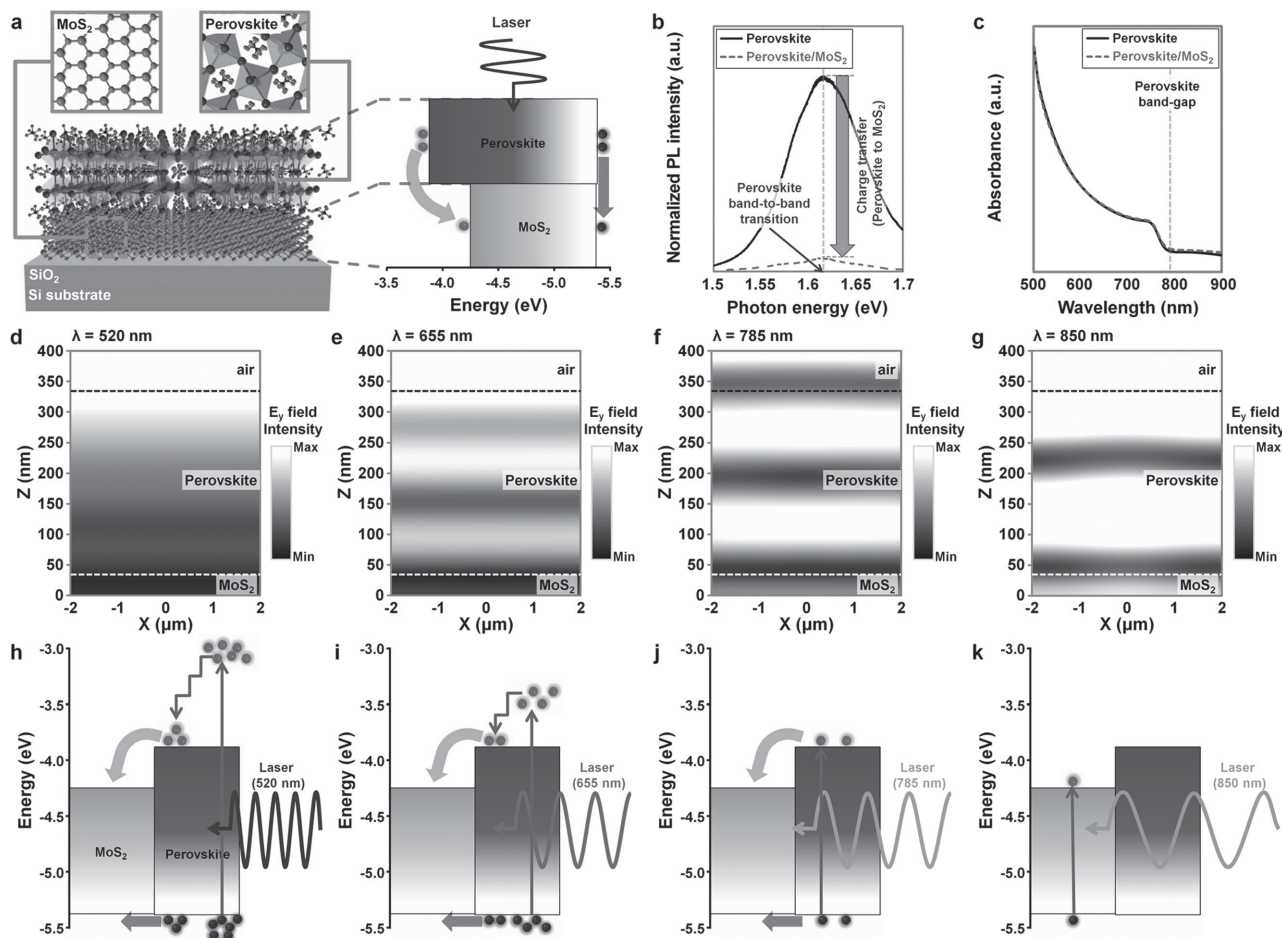


Figure 1. Characterization of the perovskite/MoS₂ hybrid structure. a) Schematic illustration and energy-band diagrams of the perovskite/MoS₂ hybrid structure. b,c) Photoluminescence (b) and UV-vis spectra (c) measured for perovskite and the perovskite/MoS₂ hybrid structure. d–k) The electric-field distribution (d–g) and energy-band diagrams (h–k) of the perovskite/MoS₂ hybrid structure under exposure to lasers with wavelengths of 520 nm (d,h), 655 nm (e,i), 785 nm (f,j), and 850 nm (g,k). Here, the electric-field distribution was estimated by RCWA optical simulation.

the perovskite/MoS₂ hybrid structure. When a laser is incident on the perovskite/MoS₂ structure, electron–hole pairs are expected to be mainly generated in the perovskite absorption layer because of its excellent absorption characteristics. Based on the good transport properties of the perovskite, such as the very long diffusion length and good ambipolar charge transport,^[29,30] the generated electrons and holes are thought to reach the MoS₂ region through diffusion. This predication is corroborated by the PL spectra obtained from perovskite-only and perovskite/MoS₂ samples (Figure 1b). Compared with the perovskite-only PL peak spectrum, a decrease in the band-to-band transition peak (at 1.61 eV) is obvious in the perovskite/MoS₂ PL spectrum, suggesting that a great portion of the photogenerated carriers was transferred from the perovskite to the MoS₂. In the case of PL spectra measured at 10 K, two peaks appear near 1.58 and 1.66 eV in both perovskite and perovskite/MoS₂ samples (see the Supporting Information, Figure S1), and a reduction in the PL peak intensity also occurs. However, it only occurs for the larger energy PL peak (at 1.66 eV) and the degree of the intensity reduction is much smaller at 10 K compared to room temperature, suggesting that the existence of a very small barrier for charge transport (from the perovskite

to the MoS₂) must be insignificant at room temperature (see the text accompanying Figure S1, Supporting Information). We then performed UV-vis analysis to investigate the characteristics of the perovskite absorption layer in the perovskite/MoS₂ hybrid structure as shown in Figure 1c. Both absorption spectra (perovskite-only and perovskite/MoS₂) are nearly identical, illustrating that most of the absorption occurs in the perovskite layer with negligible contribution from the MoS₂ layer. The bandgap of the perovskite is estimated from the near-band-edge emission at 1.61 eV from the PL, which is also similar to the apparent optical bandgap of 1.59 eV determined from the absorption spectrum (Supporting Information, Figure S2). The UV-vis absorption spectra follow an absorption curve of typical direct gap semiconductors as a function of wavelength; the number of photogenerated electron–hole pairs is expected to increase under exposure to a laser with a lower wavelength.

To investigate the absorption properties of the perovskite/MoS₂ hybrid structure in more detail, we calculated E_y-field intensity distributions under various laser wavelengths (520, 655, 785, and 850 nm) using the RCWA method. When the perovskite/MoS₂ sample is exposed to a laser with a 520 nm wavelength, most photons are expected to be absorbed in the

perovskite surface region (Figure 1d), subsequently generating a very high quantity of electron-hole pairs at the surface (Figure 1h). However, the light absorption by the perovskite decreases when a laser with a higher wavelength (lower photon energy) is used (see Figure 1c and Figure S4, Supporting Information). As a result, a laser with a 655 nm wavelength seems to reach further inside the perovskite layer (Figure 1e) and also to generate a relatively smaller number of electron-hole pairs (Figure 1i) than 520 nm laser exposure. Similar changes (deeper penetration and lower carrier generation) still appear in the perovskite/MoS₂ sample exposed to a 785 nm wavelength laser (Figures 1f,j). However, because the perovskite layer cannot absorb a laser with a wavelength above 800 nm, the 850 nm laser is thought to reach the MoS₂ region and generate electron-hole pairs only in the MoS₂ layer (Figure 1g,k).

To confirm the improvement in the optoelectronic performance of the MoS₂ photodetector through the insertion of a perovskite absorption layer, we fabricated a perovskite/MoS₂-based photodetector and investigated its optoelectronic characteristics by current-voltage (I_D - V_G) measurements under dark and illuminated conditions. Figure 2a shows the structural schematic and energy-band diagram (at $V_{DS} = 5$ V) of a perovskite-coated MoS₂ photodetector under illuminated conditions ($P_{\text{light}} = 6 \text{ mW cm}^{-2}$ and $\lambda = 520 \text{ nm}$). When the perovskite layer is exposed to this laser, electron-hole pairs are expected to be generated at the surface region (see Figure 1h). After that, the electrons and holes split based on the ambipolar charge transport property of the perovskite and diffuse into the MoS₂ region, followed by the carrier collection step caused by the electric field in the Ti-MoS₂ junctions (right side: electrons and left side: holes). The fabricated perovskite/MoS₂ photodetector

is shown in Figure 2b. As a result, as shown in Figure 2c, the photocurrent of the perovskite/MoS₂ photodetector (obtained at $V_{DS} = 5$ V) was increased by about eight times in the on-state ($V_{GS} = 20$ V), compared to the conventional MoS₂ device.

Then, to analyze the performance improvement of the perovskite/MoS₂ photodetector according to the wavelength of the exposed laser, we extracted and compared the photoresponsivity ($R = I_{\text{Photo}}/P_{\text{Light}}$) and detectivity ($D^* = (AB)^{1/2}/NEP$) values. Here, the photocurrent is $I_{\text{Laser on}} - I_{\text{Laser off}}$, P_{Light} is the total incident optical power, A is the effective detection area of the device ($25 \mu\text{m}^2$ as confirmed by the scanning electron microscopy (SEM) image in Figure 2b), B is the bandwidth of photodetector, and NEP is the noise equivalent power of the photodetector (Supporting Information, Figure S9). Figure 2d,e shows the photoresponsivity (at $V_{GS} = 20$ V) and detectivity (at $V_{GS} = -30$ V) values of the perovskite/MoS₂ photodetector and their ratios ($R_{\text{Perov}}/R_{\text{MoS}_2}$ and $D_{\text{Perov}}^*/D_{\text{MoS}_2}^*$) as a function of wavelength. In the case of the 520 nm laser, the photoresponsivity and detectivity were highly enhanced by factors of 7.7 (from 636 to $4.9 \times 10^3 \text{ A W}^{-1}$) and 5.7 (from 1.53×10^9 to 8.76×10^{10} Jones), respectively, after applying perovskite to the MoS₂ photodetector. We also observed a similar improvement in the cases of the 655 and 785 nm lasers (photoresponsivity: 1.8–2.7 and detectivity: 1.1–1.9). When the device was exposed to an 850 nm laser, however, the photoresponsivity and detectivity were not changed much because it is out of the perovskite absorption range ($R: 89.8 \rightarrow 104.24 \text{ A W}^{-1}$ and $D^*: 2.16 \times 10^{10} \rightarrow 2.39 \times 10^{10}$ Jones).

We then applied the APTES surface-treatment technique to the SiO₂/Si substrate before fabricating the MoS₂ photodetector to induce n-type doping on the MoS₂ region and eventually

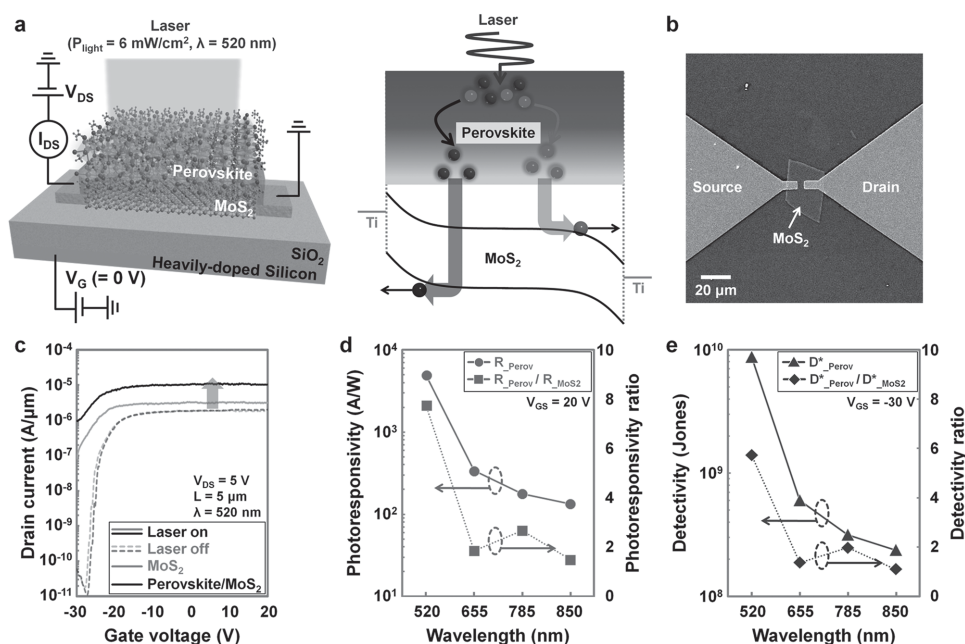


Figure 2. Optoelectronic characterization of the perovskite/MoS₂ photodetector. a) Schematic and energy-band diagram of the perovskite/MoS₂ photodetector ($V_{DS} = 5$ V and $V_{GS} = 0$ V) under laser exposure ($P_{\text{light}} = 6 \text{ mW cm}^{-2}$ and $\lambda = 520 \text{ nm}$). b) Top-view SEM image of the perovskite/MoS₂ photodetector. Here, the channel length and width were each 5 μm . c) I_D - V_G characteristics of MoS₂ (gray) and perovskite/MoS₂ (black) photodetectors under both dark (dotted line) and illuminated (solid line) conditions. d,e) Photoresponsivity (d) and detectivity (e) values of perovskite/MoS₂ photodetectors as a function of the laser wavelength. Here, the photoresponsivity and detectivity values were extracted at $V_{GS} = 20$ V and $V_{GS} = -30$ V, respectively.

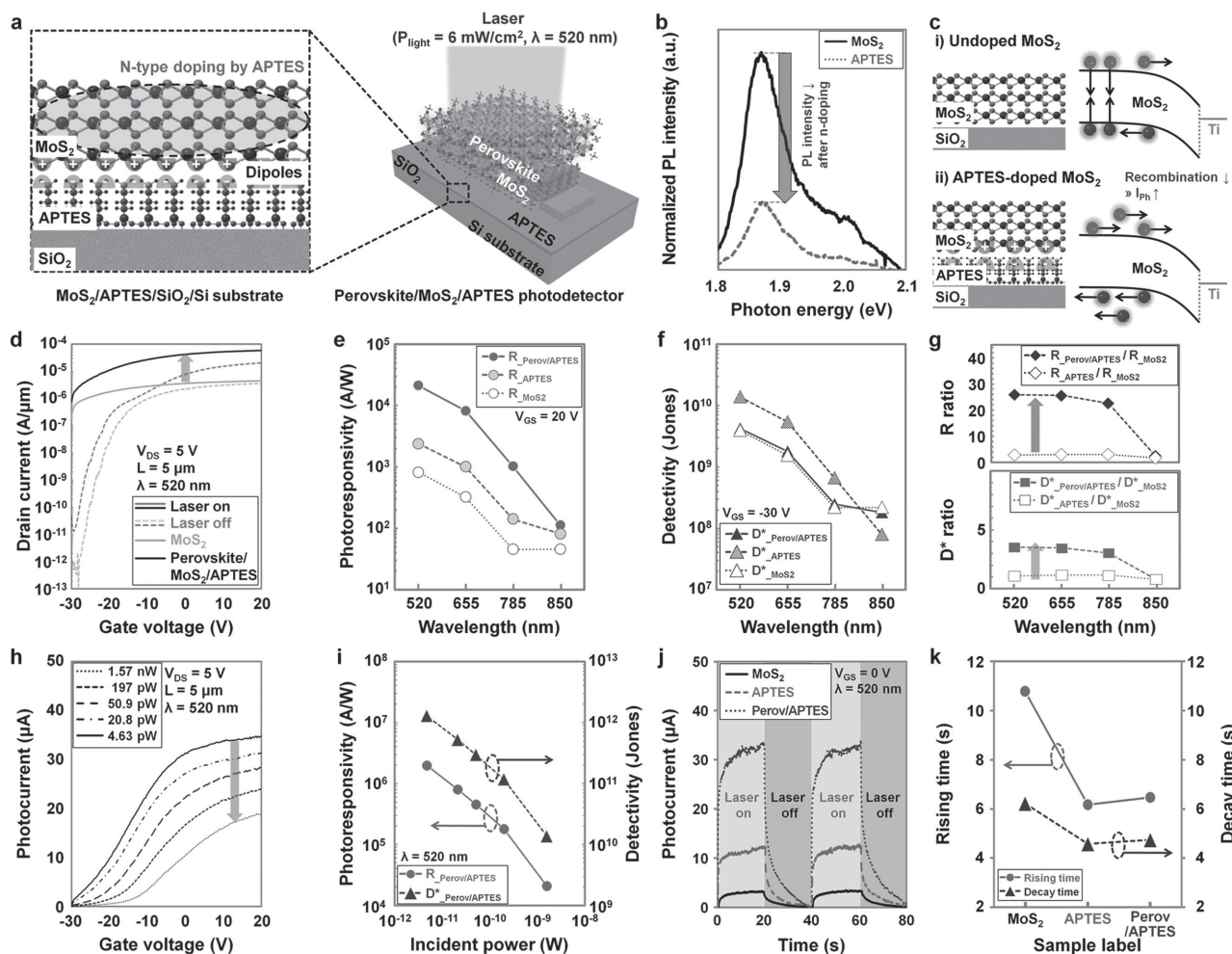


Figure 3. Characterization of the APTES-doped perovskite/MoS₂ photodetector. a) Schematic diagram of the perovskite/MoS₂/APTES photodetector under laser exposure and cross-sectional structure explaining the APTES-doping mechanism on the MoS₂ film. b) Photoluminescence spectra ($T = 200$ K) of MoS₂ and MoS₂/APTES samples. c) Schematic energy-band diagrams showing the APTES-doping effects in the MoS₂-Ti junction and MoS₂ region. d) I_D - V_G characteristics of MoS₂ (gray) and perovskite/MoS₂/APTES (black) photodetectors under both dark (dotted line) and illuminated (solid line) conditions. e-g) Photoresponsivity (e), detectivity (f), and their ratio (g) to the MoS₂ as a function of wavelength in MoS₂, MoS₂/APTES, and perovskite/MoS₂/APTES photodetectors. h) I_D - V_G characteristics of the perovskite/MoS₂/APTES photodetector according to the optical power of the 520-nm incident laser. i) Photoresponsivity and detectivity values as a function of incident laser power. j) Photoswitching characteristics ($V_{GS} = 0$ V and $\lambda = 520$ nm), and k) rising/decay times of MoS₂, MoS₂/APTES, and perovskite/MoS₂/APTES photodetectors.

create a synergy effect with perovskite in terms of photodetector performance. **Figure 3a** explains the n-type doping mechanism of MoS₂ by APTES, where a functional-group amino ($-NH_2$) of the APTES has a negative pole, subsequently reducing the number of hole carriers in the MoS₂ (n-type doping phenomenon).^[31,32] We first performed PL analysis on the undoped/APTES-doped MoS₂ films, as shown in **Figure 3b**, and a reduction in MoS₂ PL peaks was observed through the APTES doping. This change in PL peak is thought to subsequently reduce the recombination rate of photogenerated charges. As a result, the APTES-doped MoS₂ photodetector collected larger amounts of photogenerated carriers than the undoped sample, thereby increasing the photocurrent of the MoS₂ photodetector (see the schematic diagram in **Figure 3c**). **Figure 3d** shows the I_D - V_G characteristics of the MoS₂ and perovskite/MoS₂/APTES photodetectors under dark and illuminated conditions. Compared with the MoS₂ sample, the photocurrent level of the

perovskite/MoS₂/APTES device increased dramatically, by a factor of 25.9 (at $V_{GS} = 20$ V and $\lambda = 520$ nm). This improvement also outstripped the perovskite/MoS₂ device without APTES doping (7.7, **Figure 2c**), obviously presenting a synergy effect for perovskite absorption and APTES n-doping layers. We then obtained the photoresponsivity and detectivity values as a function of the wavelength of the incident laser from the MoS₂, perovskite/MoS₂, and perovskite/MoS₂/APTES devices, as shown in **Figure 3e,f**. In the case of 520 nm laser exposure, the photoresponsivity and detectivity of the perovskite/MoS₂/APTES device were improved by factors of 25.9 ($R: 816.6 \rightarrow 2.11 \times 10^4$ A W⁻¹) and 3.5 ($D^*: 3.93 \times 10^9 \rightarrow 1.38 \times 10^{10}$ Jones), respectively, where the values were much higher than those of the MoS₂/APTES ($R_{APTES}/R_{MoS_2}: 2.91$ and $D^*_{APTES}/D^*_{MoS_2}: 1.08$) or perovskite/MoS₂ ($R_{Perov}/R_{MoS_2}: 7.7$ and $D^*_{Perov}/D^*_{MoS_2}: 5.7$) devices. In the cases of the 655 and 785 nm wavelengths, the photoresponsivity and detectivity were also increased, presenting a

Table 1. Performance summary of all devices under the exposure to lasers with various wavelengths.

Parameters	Wavelength [nm]	Structure		
		MoS ₂	MoS ₂ /APTES	Perovskite/ MoS ₂ /APTES
Photoresponsivity [A W ⁻¹] at V _{GS} = 20 V	520	8.16 × 10 ²	2.38 × 10 ³	2.12 × 10 ⁴
	655	3.21 × 10 ²	1.01 × 10 ³	8.24 × 10 ³
	785	4.52 × 10 ¹	1.41 × 10 ²	1.02 × 10 ³
	850	4.55 × 10 ¹	8.14 × 10 ¹	1.11 × 10 ²
	520	3.93 × 10 ⁹	4.23 × 10 ⁹	1.38 × 10 ¹⁰
Detectivity [Jones] at V _{GS} = -30 V	655	1.55 × 10 ⁹	1.79 × 10 ⁹	5.37 × 10 ⁹
	785	2.18 × 10 ⁸	2.51 × 10 ⁸	6.66 × 10 ⁸
	850	2.19 × 10 ⁸	1.80 × 10 ⁸	7.93 × 10 ⁷
	520	3.93 × 10 ⁹	4.23 × 10 ⁹	1.38 × 10 ¹⁰

ratio of 22.57–25.62 for photoresponsivity and 3.05–3.47 for detectivity. On the other hand, under 850 nm laser exposure, no performance improvement was obtained in any of the devices because it is out of the absorption range of perovskite. The performance results (photoresponsivity and detectivity) of all the devices under the exposure of lasers at various wavelengths are summarized in **Table 1**.

In addition, we performed extended analysis of the perovskite/MoS₂/APTES photodetector in terms of incident power (Figure 3h,i) and temporal photoresponse (Figure 3j,k). Figure 3h shows the I_D - V_G characteristics according to the incident power (between 1.57 nW and 4.63 pW) under 520 nm laser exposure. In general, photoresponsivity depends on the power of the incident laser (for reference, $R \propto 1/P_{\text{light}}$). However, in the proposed perovskite/MoS₂/APTES photodetector, the photoresponsivity exponentially increases as the incident power decreases because of the reduced scattering between the photoexcited electrons.^[32] Figure 3i shows the photoresponsivity and detectivity values as a function of the power of the incident laser. The photoresponsivity value was enhanced by 94.2-fold (from 2.11×10^4 to 1.94×10^6 A W⁻¹), which is extremely high compared with that of other TMD-based photodetectors (1.45×10^4 A W⁻¹ in OTS-doped WSe₂^[32] and 6×10^5 A W⁻¹ in PbS/MoS₂^[24]) and perovskite-applied photodetectors (180 A W⁻¹ in perovskite/graphene^[23] and 242 A W⁻¹ in an organolead-triiodide-perovskite-based photodetector.^[33]). The detectivity showed a similar tendency to depend on the power of the incident laser ($D^* \propto R$). The obtained highest detectivity value in this device (1.29×10^{12} Jones) was comparable to the other reported TMD-based photodetectors (2.2×10^{12} Jones in ferroelectric/MoS₂^[34] and 1.18×10^{12} Jones in few-layer ReS₂^[35]). In particular, when the gate bias was fixed at -30 V, 9.16×10^4 A W⁻¹ photoresponsivity and 1.29×10^{12} Jones detectivity were simultaneously obtained for the Perov/MoS₂/APTES photodetector. This performance was much better than that for photodetectors previously formed on conventional TMDs (see the Supporting Information, Table S2) and also comparable to that of the ReS₂-based photodetector that showed the best performance up to now ($R = 8.8 \times 10^4$ A W⁻¹ and $D^* = 1.182 \times 10^{12}$ Jones).^[35] In addition, as shown in Figure 3j, the

photoswitching characteristics of the MoS₂, MoS₂/APTES, and perovskite/MoS₂/APTES photodetectors were investigated under a 520 nm laser that turned on and off on a 20 s cycle. From the temporal photoresponse curves, we then extracted rising and decay time values (Figure 3k). Here, all the curves were fitted to the exponential function (see the Supporting Information, Figure S5). In the MoS₂ sample, the rising and decay time values were 10.7 and 6.2 s, respectively, but the slow temporal photoresponse was improved by APTES doping (rising time: 6.17 s, decay time: 4.5 s) because: i) the insertion of APTES prevented the increase of the carrier recombination rate by the MoS₂-SiO₂ interfacial traps and ii) the n-doping reduced the recombination rate in the MoS₂ region (which is a transporting layer). Recently, Lopez-Sanchez et al. reported a reduction in the photoresponsivity by the trap states present either in the MoS₂ or at the interface between the MoS₂ and the underlying SiO₂ layer.^[11] Based on additional experiments performed with WSe₂, shown in Figure S7, we also note that the slow speed problem of the perovskite/MoS₂ photodetector can be resolved by selecting an alternative TMD material.

Meanwhile, perovskite films have poor stability in air because of their high reactivity with moisture.^[36] Although single-crystal perovskite film is known to have a good air stability,^[37] it is necessary to encapsulate the perovskite/MoS₂ and perovskite/MoS₂/APTES photodetectors using optically transparent layers. To improve the stability of the perovskite-based TMD photodetectors, we therefore adopted a bilayer of OTS and PMMA to encapsulate the photodetectors. We then monitored the device photoresponsivity and detectivity as a function of their air-exposure time. The OTS on PMMA is expected to enhance the protection effect of PMMA by forming a more hydrophobic surface. **Figure 4a** shows cross-sectional schematic diagrams of: i) nonencapsulated, ii) PMMA-covered, and iii) OTS/PMMA-covered perovskite/MoS₂/APTES photodetectors and their corresponding contact angles, which were measured on the surface of each sample. The contact angle of the pristine perovskite film was about 52.5°, which would allow moisture to be easily attached on the perovskite surface and react with perovskite. However, the contact angle was increased by coating the perovskite with a PMMA layer (71.5°) and an OTS/PMMA bilayer (107°), demonstrating the feasibility of an OTS/PMMA bilayer to protect perovskite-based TMD photodetectors. We then monitored the photoresponsivity and detectivity of the three kinds of TMD photodetectors as a function of air-exposure time, as shown in Figure 4b,c. In the non-encapsulated device, both the photoresponsivity and detectivity deteriorated completely after 48 h in air. Although the degree of performance degradation was somewhat retarded in the PMMA-covered device, it was still unstable in air (deterioration of about 50% in photoresponsivity and 70% in detectivity after 68 h of air exposure). However, the OTS/PMMA-covered device showed better stability (deterioration of about 20.4% in photoresponsivity and 28.7% in detectivity after 200 h). The perovskite/MoS₂/APTES devices presented similar stability under the illumination of lasers with various wavelengths (degradation of 20.4–24.8% in photoresponsivity and 28.7–34.2% in detectivity), as shown in Figure 4d,e. This performance degradation observed in the devices encapsulated by the PMMA/OTS layer seems to be attributed to the inflow of moisture for less than 60 s before the

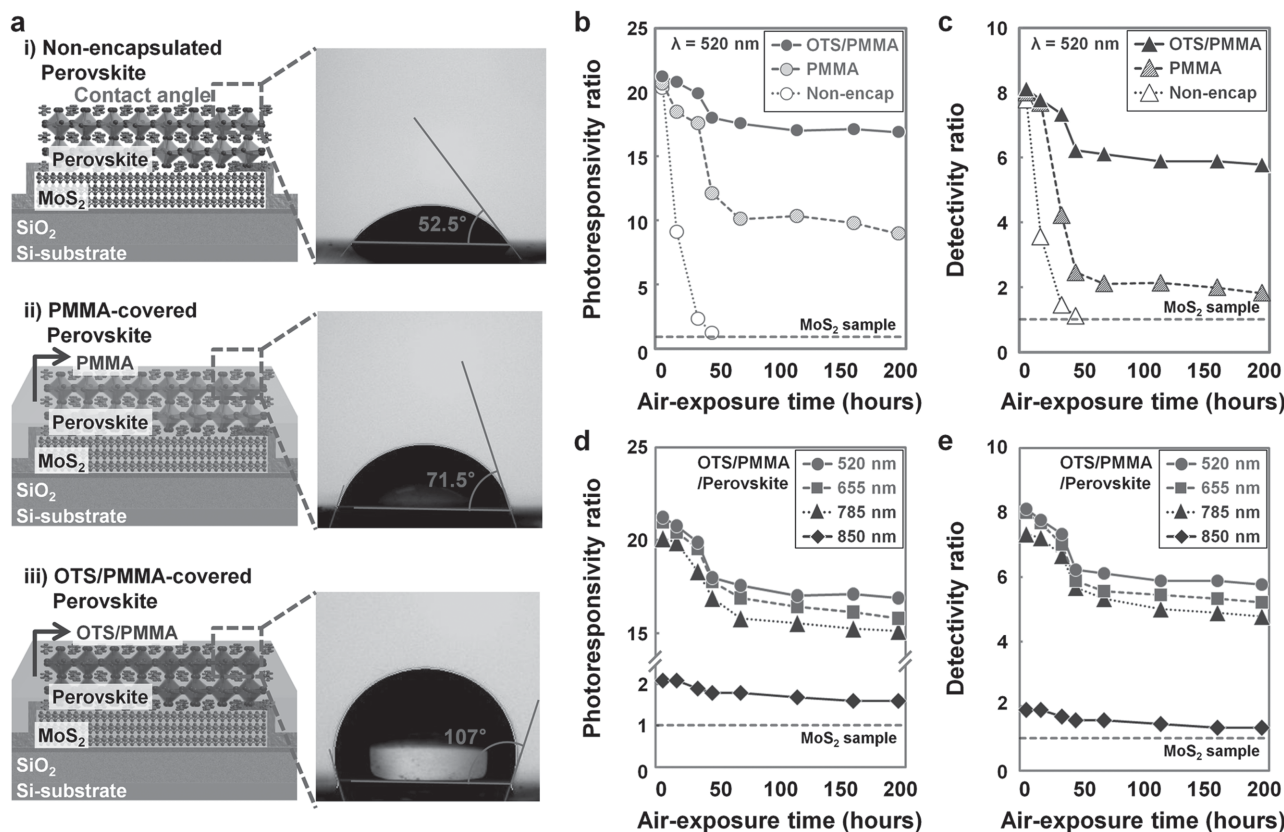


Figure 4. Stability of perovskite/MoS₂/APTES photodetectors in air. a) Cross-sectional schematic diagrams and contact angles of: i) nonencapsulated, ii) PMMA-covered, and iii) OTS/PMMA-covered perovskite/MoS₂/APTES photodetectors. Here, the contact angle was measured on the sample surfaces. b,c) Photoresponsivity (b) and detectivity ratio (c) of the three kinds of perovskite/MoS₂/APTES photodetectors as a function of air-exposure time ($\lambda = 520$ nm). d,e) Photoresponsivity (d) and detectivity ratio (e) of OTS/PMMA-covered perovskite/MoS₂/APTES photodetector as a function of air-exposure time according to the laser wavelengths (520, 655, 785, and 850 nm).

encapsulation process is completed. Although perovskite film was formed in a N₂ ambient (glove box), the encapsulation process was performed in air, probably causing the decomposition of perovskite film and the device degradation.

Through PL and UV-vis analyses of the perovskite/MoS₂ samples, we first confirmed that most of the photogenerated carriers were transferred from the perovskite to the MoS₂. In particular, we investigated the absorption properties of the perovskite/MoS₂ hybrid structure in detail using the RCWA method. In the fabricated perovskite/MoS₂ photodetector, the photoresponsivity and detectivity were enhanced from 636 to 4.9×10^3 A W⁻¹ and from 1.53×10^9 to 8.76×10^9 Jones, respectively, under 520 nm laser exposure. Then, by applying APTES n-doping on the MoS₂ region, we created a synergy effect with perovskite in terms of the photodetector performance (photoresponsivity and detectivity values of the perovskite/MoS₂/APTES device were improved from 816.6 to 2.11×10^4 A W⁻¹ (R) and from 3.93×10^9 to 1.38×10^{10} Jones (D^*)). In addition, in the perovskite/MoS₂/APTES photodetector, the photocurrent exponentially increased as the incident power decreased because of reduced scattering between the photoexcited electrons, presenting the highest photoresponsivity and detectivity values ($R = 1.94 \times 10^6$ A W⁻¹ and $D^* = 1.29 \times 10^{12}$ Jones). We also minimized the performance degradation caused by moisture down to 20% in the

perovskite/MoS₂/APTES photodetector by adopting a new encapsulation bilayer of OTS and PMMA.

Experimental Section

Formation of Perovskite (CH₃NH₃PbI₃) Film: A 400 mg mL⁻¹ solution of PbI₂ (Sigma-Aldrich) dissolved in *N,N*-dimethylformamide (DMF) (Sigma-Aldrich) was spin-coated onto the substrate at 8000 rpm for 30 s (8 s acceleration). Then, a 30 mg mL⁻¹ solution of CH₃NH₃I (Dyesol) dissolved in isopropyl alcohol was sequentially loaded onto the PbI₂ layer for 20 s. After spinning at 8000 rpm for 30 s, the samples were annealed at 100 °C for 5 h. Here, the thickness of perovskite film was about 320 nm (Supporting Information, Figure S6).

Preparation of MoS₂ Flakes on the SiO₂/Si Substrate: MoS₂ flakes (2D semiconductor) were mechanically exfoliated and transferred onto the 90 nm thick SiO₂/Si substrate using adhesive tape. Here, the thickness of the MoS₂ flake was about 25 nm (Supporting Information, Figure S6). Then, an acetone wash was performed on the samples for 1 h to remove the tape residue.

Photoluminescence, UV-vis, and Absorption Measurements: Steady-state PL measurements at room temperature and 10 K were performed using a labRAM HR UV/Vis/NIR PL (Horiba Jobin Yvon, France) using a 514 nm argon laser excitation source (a spot size of 1–2 μ m) and a charge-coupled device (CCD) detector. The intensity of the source was varied from 17 to 0.17 mW cm⁻². In addition, absorption measurements were carried out using a Lambda 1050 between 200 and 1000 nm wavelengths at room temperature (a deuterium and tungsten

lamp excitation source was used together with a photomultiplier tube, InGaAs, and PbS detector).

Optical Simulation for Electric-Field Distributions: To investigate the absorption profile in constituent materials (i.e., perovskite and MoS₂) for penetrating light, the electric-field distributions of the perovskite/TMD hybrid structure were calculated at wavelengths of 520, 655, 785, and 850 nm by the RCWA method, which is based on Maxwell's equations in the Fourier domain, using commercial software (*DiffractionMOD*, Rsoft Design Group). In this simulation, it was assumed that linearly polarized light (i.e., TE wave) enters the perovskite/MoS₂ hybrid structure from the air. The thicknesses of the perovskite and MoS₂ layers were set to 300 and 30 nm, respectively. The optical parameters (i.e., n and k) of the perovskite and MoS₂ were acquired from previous reports.^[38,39]

Fabrication and Optical Measurements of MoS₂ and Perovskite/MoS₂ Photodetectors: For the fabrication of a back-gated MoS₂ photodetector, the source/drain electrode regions (5 μm of channel length and width) were patterned on the MoS₂/SiO₂/Si sample by optical lithography, followed by Ti (10 nm) and Au (50 nm) deposition via an e-beam evaporator. Then, the perovskite absorbing layer was formed by a spin-coating process on the MoS₂ photodetector. The fabricated photodetectors were analyzed by current–voltage (I_D – V_G) measurements using an HP 4155A semiconductor parameter analyzer under both dark and illuminated conditions. The light source was a dot laser with wavelengths of 520, 655, 785, and 850 nm, where the optical power was about 6 mW cm^{−2}. To characterize and compare the optoelectronic performance of MoS₂ and perovskite/MoS₂ photodetectors, the photoresponsivity (R) and detectivity (D^*) were calculated from the I_D – V_G . R is $I_{\text{photo}}/P_{\text{light}}$ and D^* is $(AB)^{1/2}/NEP$, where I_{photo} is the generated photocurrent ($I_{\text{Laser on}} - I_{\text{Laser off}}$), P_{light} is the total incident optical power, A is the effective area of the detector, B is the bandwidth of the photodetector (0.1 Hz), and NEP is the noise equivalent power of the photodetector (Supporting Information, Figure S9).

APTES Surface-Treatment Technique for n-Doping of MoS₂ Film: For the formation of the APTES layer on the SiO₂/Si surface, 0.1 mL of APTES was added to 50 mL of toluene and the SiO₂/Si substrate was soaked in the solution for 1 h. After that, the substrate was rinsed with toluene, alcohol, and deionized (DI) water by sonication (for 10 min in each step). To achieve APTES n-doping of MoS₂, the fabricated MoS₂ devices were spin-coated with PMMA at 3000 rpm and baked at 130 °C for 2 min. Then the PMMA-covered samples were floated on 2% hydrogen fluoride (HF) and deionized water (wet-transfer technique) and transferred onto the APTES-treated SiO₂/Si substrate. After that, the PMMA was removed by acetone wash, and the MoS₂/APTES samples were baked at 120 °C for 20 min.

Encapsulation of Perovskite/MoS₂/APTES Photodetector: To encapsulate the perovskite/MoS₂/APTES photodetector, the PMMA solution was spin-coated on the devices at 3000 rpm and baked at 130 °C for 2 min. The OTS solution was then dropped and dried on the PMMA-coated devices for 5 min and then rinsed several times with toluene and DI water.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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